📊 SLI, SLO & SLA — Easy KCNA Notes

These three terms are very important in **Cloud Native, DevOps, and Site Reliability Engineering (SRE)**.

The easiest way to remember them is:

SLI = What do we measure?
SLO = What target do we want?
SLA = What promise do we make to the customer?

---

1. SLI — Service Level Indicator 📊

Simple definition

SLI is a measurement of how well a service is performing.

It is the **actual data/measurement**.

Examples

You can measure:

- Availability
- Latency
- Error rate
- Response time
- Throughput

For example:

Website availability = 99.95%

That **99.95%** is an SLI measurement.

🧠 **Easy memory:**

SLI = Measurement

---

2. SLO — Service Level Objective 🎯

Simple definition

SLO is the target level of performance that you want your service to achieve.

For example:

SLO:

Website availability should be ≥ 99.9%

Your actual measurement might be:

SLI = 99.95%

SLO = 99.9%

So your service is meeting the target.

🧠 **Easy memory:**

SLO = Target

---

3. SLA — Service Level Agreement 🤝

Simple definition

SLA is a formal agreement between a service provider and a customer about the expected level of service.

For example, a cloud provider might promise:

Availability = 99.9%

If the provider fails to meet the agreement, the SLA may specify consequences such as **service credits**, depending on the contract.

🧠 **Easy memory:**

SLA = Promise/Agreement

---

4. Simple Real-Life Example 🏫

Imagine a school.

SLI 📊 — Measurement

You measure:

Student attendance = 95%

That's the actual measurement.

SLO 🎯 — Target

The school wants:

Attendance target = 90%

That's the target.

SLA 🤝 — Agreement

The school makes a formal commitment:

"We will maintain at least 90% attendance."

That's the agreement/commitment.

---

5. How SLI, SLO and SLA Work Together ⭐⭐⭐

Think of them as a chain:

SLI

↓

Actual Measurement

↓

SLO

↓

Target

↓

SLA

↓

Customer Agreement

Example:

SLI = 99.95% availability

↓

SLO = 99.9% availability

↓

SLA = Provider promises 99.9% availability

---

6. Another Example: Website Response Time 🌐

Suppose you run an online shopping website.

SLI

You measure actual response time:

Average response time = 180 ms

SLO

You set a target:

Response time should be < 200 ms

SLA

You promise the customer:

99% of requests will meet the agreed response-time requirement.

So:

📊 SLI → What actually happened?

🎯 SLO → What target did we set?

🤝 SLA → What did we promise?

---

7. SLI vs SLO vs SLA ⭐

Term	Full Form	Easy Meaning
SLI	Service Level Indicator	📊 Measurement
SLO	Service Level Objective	🎯 Target

Term	Full Form	Easy Meaning
SLA	Service Level Agreement	🤝 Customer agreement

---

8. Important Difference Between SLO and SLA

Don't confuse these two.

SLO

An **internal target/objective** for how the service should perform.

"We want 99.9% availability."

SLA

A **formal commitment/agreement** with customers.

"We promise 99.9% availability under this agreement."

An SLA may include consequences if the agreed level isn't met.

🧠 **Remember:**

SLO = Goal

SLA = Contract/Promise

---

9. Error Budget ⭐

You may hear this term together with SLO.

Suppose:

SLO = 99.9% availability

That means you allow:

100% - 99.9% = 0.1%

of unavailability.

That allowed amount is called the **error budget**.

Simple idea:

SLO = 99.9%

↓

Allowed failure = 0.1%

↓

Error Budget

The error budget helps teams balance **reliability** with **releasing new features**.

🧠 KCNA Revision Notes

SLI 📊

- Actual measurement of service performance.
- Examples: availability, latency, error rate.
- **SLI = What are we measuring?**

SLO 🎯

- Desired performance target.
- Example: availability ≥ 99.9%.
- **SLO = What target do we want?**

SLA 🤝

- Formal agreement with the customer.
- Defines expected service level and potentially consequences if it isn't met.
- **SLA = What do we promise?**

⭐ Super Easy Memory Trick

SLI → 📊 MEASURE

SLO → 🎯 TARGET

SLA → 🤝 PROMISE

One-line exam answer:

SLI measures service performance, SLO defines the desired target, and SLA is the formal agreement with the customer about the expected service level.